

Learning Scalar Adjective Intensity from Paraphrases

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Abstract

Adjectives like *warm*, *hot*, and *scalding* all describe temperature but differ in intensity. Understanding these differences between adjectives is a necessary part of reasoning about natural language. We propose a new paraphrase-based method to automatically learn the relative intensity relation that holds between a pair of scalar adjectives. Our approach analyzes over 36k adjectival pairs from the Paraphrase Database under the assumption that, for example, paraphrase pair *really hot* $\leftrightarrow$ *scalding* suggests that *hot* $<$ *scalding*. We show that combining this paraphrase evidence with existing, complementary pattern- and lexicon-based approaches improves the quality of systems for automatically ordering sets of scalar adjectives and inferring the polarity of indirect answers to *yes/no* questions.

1 Introduction

Semantically similar adjectives are not fully interchangeable in context. Although *hot* and *scalding* are related, the statement “*the coffee was hot*” does not imply the coffee was *scalding*. *Hot* and *scalding* are scalar adjectives that describe *temperature*, but they are not interchangeable because they vary in intensity. A native English speaker knows that their relative intensities are given by the ranking *hot* $<$ *scalding*. Understanding this distinction is important for language understanding tasks such as sentiment analysis (Pang et al., 2008), question answering (de Marneffe et al., 2010), and textual inference (Dagan et al., 2006).

Existing lexical resources such as WordNet (Miller, 1995; Fellbaum, 1998) do not include the relative intensities of adjectives. As a result, there have been efforts to automate the process of learning intensity relations (e.g. Sheinman and Tokunaga (2009), de Melo and Bansal (2013), Wilkinson (2017), etc.). Many existing approaches rely

<i>particularly pleased</i>	$\leftrightarrow$	<i>ecstatic</i>
<i>quite limited</i>	$\leftrightarrow$	<i>restricted</i>
<i>rather odd</i>	$\leftrightarrow$	<i>crazy</i>
<i>so silly</i>	$\leftrightarrow$	<i>dumb</i>
<i>completely mad</i>	$\leftrightarrow$	<i>crazy</i>

Figure 1: Examples of paraphrases from PPDB of the form $RB\ JJ_u \leftrightarrow JJ_v$ which can be used to infer pairwise intensity relationships ($JJ_u < JJ_v$).

on *pattern-based* or *lexicon-based* methods to predict the intensity ranking of adjectives. Pattern-based approaches search large corpora for lexical patterns that indicate an intensity relationship – for example, “*not just X, but Y*” implies $X < Y$. As with pattern-based approaches for other tasks (such as hypernym discovery (Hearst, 1992)), they are precise but have relatively sparse coverage of comparable adjectives, even when using web-scale corpora (de Melo and Bansal, 2013; Ruppenhofer et al., 2014). Lexicon-based approaches employ resources that map an adjective to a real-valued number that encodes both intensity and polarity (e.g. *good* might map to 1 and *phenomenal* to 5, while *bad* maps to -1 and *awful* to -3). They can also be precise, but may not cover all adjectives of interest.

We propose paraphrases as a new source of evidence for the relative intensity of scalar adjectives. A paraphrase is a pair of words or phrases with approximately similar meaning, such as *really great* $\leftrightarrow$ *phenomenal*. Adjectival paraphrases can be exploited to uncover intensity relationships. A paraphrase pair of the above form, where one phrase is composed of an intensifying adverb and an adjective (*really great*) and the other is a single-word adjective (*phenomenal*), provides evidence that *great* $<$ *phenomenal*. By drawing this evidence from large, automatically-generated